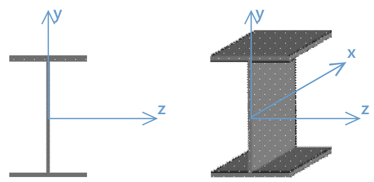
Detail Report: M1

Load Combination: LC 2: ASCE ASD 2 **Code check:** 0.732 (axial/bending)



Input Data

Shape:	W8X21	**I Node:**	N1
Member Type:	Beam	**J Node:**	N3
Length (ft):	19.25	**I Release:**	Fixed
Material Type:	Hot Rolled Steel	**J Release:**	Fixed
Design Rule:	Typical	**I Offset:**	N/A
Internal Sections:	97	**J Offset:**	N/A
Design Code:	AISC 16th (360-22): ASD	**T/C Only:**	Both Way

Material Properties

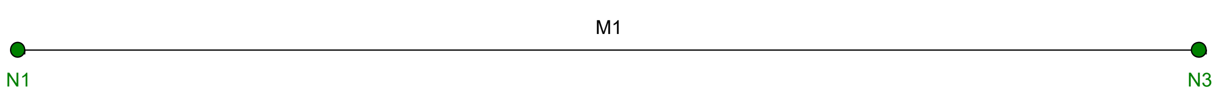
Material:	A992	**Therm. Coeff. (/1E5 F):**	0.65	**F_u (ksi):**	65
E (ksi):	29000	**Density (k/ft^3):**	0.49	**R_t:**	1.1
G (ksi):	11154	**F_y (ksi):**	50		
Nu:	0.3	**R_y:**	1.1		

Shape Properties

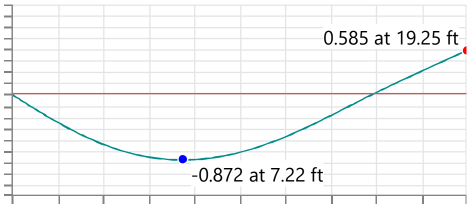
d (in):	8.28	**Area (in^2):**	6.16	**r_T (in):**	1.41
b_f (in):	5.27	**Z_{yy} (in^3):**	5.69	**J (in^4):**	0.282
t_f (in):	0.4	**Z_{zz} (in^3):**	20.4	**k_{det} (in):**	0.875
t_w (in):	0.25	**C_w (in^6):**	152	**k_{des} (in):**	0.7
I_{yy} (in^4):	9.77	**W_{no} (in^2):**	10.4		
I_{zz} (in^4):	75.3	**S_w (in^4):**	5.47		

Design Properties

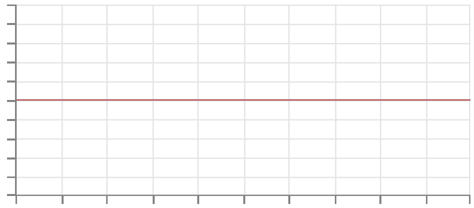
$L_{b\ y\text{-}y}$ (ft):	0	**$K_{y\text{-}y}$:**	1	**Seismic DR:**	None
$L_{b\ z\text{-}z}$ (ft):	0	**$K_{z\text{-}z}$:**	1	**Max Defl Ratio:**	L/164
$L_{comp\ top}$ (ft):	0	**y sway:**	No	**Max Defl Location:**	19.25
$L_{comp\ bot}$ (ft):	0	**z sway:**	No	**Span:**	2
L_{torque} (ft):	0	**Function:**	Lateral	**τ_b:**	1

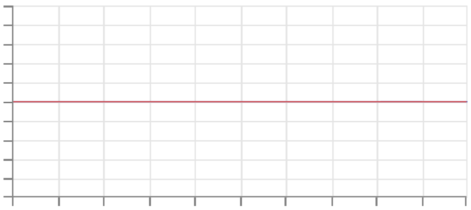


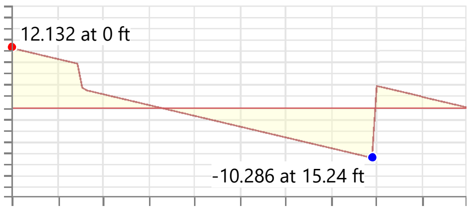
Diagrams:



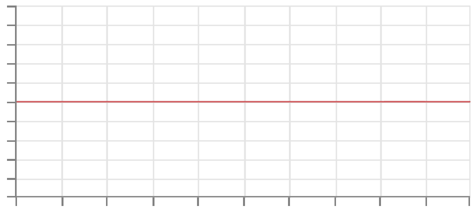
y Deflection (in)

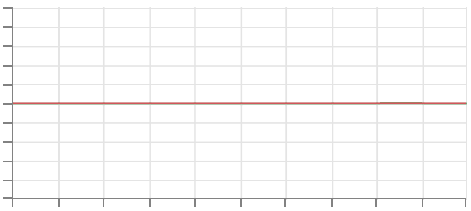

z Deflection (in)

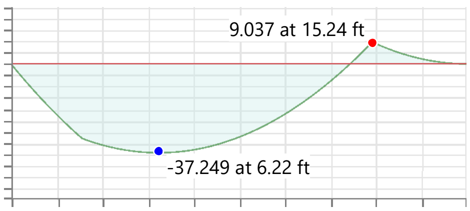

Axial Force (k)



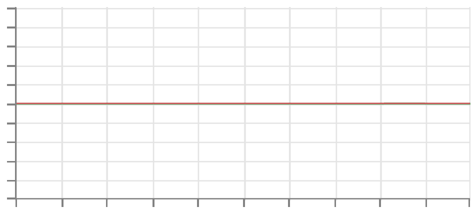
y Shear Force (k)

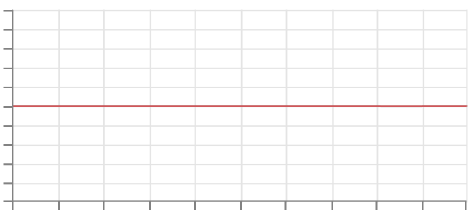

z Shear Force (k)

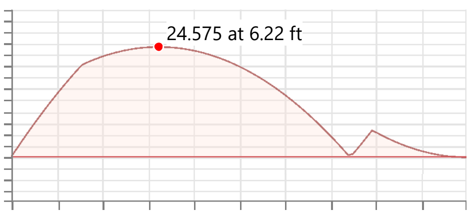

Torsion (k-ft)



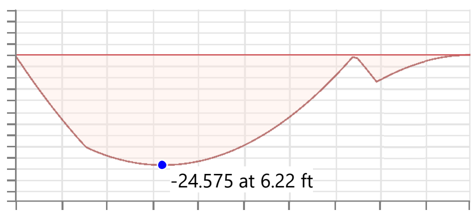
z-z Moment (k-ft)


y-y Moment (k-ft)


Axial Stress (ksi)



Bending Compression Stress (ksi)



Bending Tension Stress (ksi)

AISC 16th (360-22): ASD Code Check

Limit State		Required	Available	Unity Check	Result
Applied Loading - Bending/Axial		-	-	-	-
$Location = 6.216\ ft$	Governing bending and axial location				
$\sigma_{by,top} = 0\ ksi$	Extreme top fiber positive bending stress about y-axis				
$\sigma_{by,bot} = 0\ ksi$	Extreme bottom fiber positive bending stress about y-axis				
$\sigma_{bz,top} = 24.575\ ksi$	Extreme top fiber positive bending stress about z-axis				
$\sigma_{bz,bot} = -24.575\ ksi$	Extreme bottom fiber positive bending stress about z-axis				
$\sigma_{wy,top} = 0\ ksi$	Local top warping bending stress (per member torsion spreadsheet) at governing location				
$\sigma_{wy,bot} = 0\ ksi$	Local bottom warping bending stress (per member torsion spreadsheet) at governing location				

$\sigma_{bz,comp} = \sigma_{bz,top} = 24.575\ ksi$	Compressive Stress from bending about z-axis				
$\sigma_{bz,ten} = -\sigma_{bz,bot} = 24.575\ ksi$	Tensile stress from bending about z-axis				
$\sigma_{by,comp} = \sigma_{wy,top} = 0\ ksi$					
$\sigma_{by,ten} = -\sigma_{by,bot} + \sigma_{wy,bot} = 0\ ksi$	Tensile stress from bending about y-axis (include warping)				
$P_r(comp) = 0\ k$	Required axial compressive strength at governing location				
$S_z = 18.188\ in^3$	Elastic section modulus about local z-axis (strong)				
$S_y = 3.708\ in^3$	Elastic section modulus about local y-axis (weak)				
$M_{rz} = \sigma_{bz,comp} S_z = 37.249\ k\text{-}ft$	Required flexural strength about z-axis at governing location				
$M_{ry} = \sigma_{by,comp} S_y = 0\ k\text{-}ft$	Required flexural strength about y-axis at governing location				
Applied Loading - Shear + Torsion		-	-	-	-
$Location = 0\ ft$	Governing shear location per governing axis				
$\sigma_{vy} = 5.861\ ksi$	Shear stress in the y-axis				
$\sigma_{vz} = 0\ ksi$	Shear stress in the z-axis				
$\tau = 0\ ksi$	Shear stress due to torsion				
$\sigma_{wz} = 0\ ksi$	Warping stress in the z-axis due to torsion				
$A_{wy} = \frac{A_{vy}}{1.0} = 2.07\ in^2$	Effective shear area in local y-axis for required strength				
$V_{ry} = (\lvert\sigma_{vy}\rvert + \lvert\tau\rvert)A_{wy} = 12.132\ k$	Required shear strength about the y-axis at governing location				
$A_{wz} = \frac{A_{vz}}{1.0} = 4.216\ in^2$	Effective shear area in local z-axis for required strength				
$V_{rz} = (\lvert\sigma_{vz}\rvert + \lvert\tau\rvert + \lvert\sigma_{wz}\rvert)A_{wz} = 0\ k$	Required shear strength about the z-axis at governing location				
Axial Tension Analysis		0 k	184.431 k	-	-
Check Available Tensile Strength					
$A_g = 6.16\ in^2$	Gross cross-sectional area				
$F_y = 50\ ksi$	Specified minimum yield stress				
$\Omega_t = 1.67$	Safety factor for tension				
$P_n = F_y A_g = 308\ k$	Nominal axial strength in tension			(Eq. D2-1)	
$P_{nt} = \frac{P_n}{\Omega_t} = 184.431\ k$	Available tensile strength			(Sec. D2)	
Axial Compression Analysis		0 k	184.431 k	-	-
Check Slenderness Ratio: Compression					
$F_y = 50\ ksi$	Specified minimum yield stress				
$E = 29000\ ksi$	Modulus of elasticity				
$t_w = 0.25\ in$	Thickness of web				
$t_f = 0.4\ in$	Thickness of flange				
$b = \frac{b_f}{2} = 2.635\ in$	Half the full flange width				
$k_{des} = 0.7\ in$	Flange thickness plus fillet depth				

$h = d - 2k_{des} = 6.88\ in$	Clear depth of web inbetween flanges	
$\lambda_f = \frac{b}{t_f} = 6.588$	Flange slenderness ratio	
$\lambda_r = 0.56\sqrt{\frac{E}{F_y}} = 13.487$	Limiting width to thickness parameter for a slender element subject to axial compression	(Table B4.1a, Case 1)
$\lambda_f < \lambda_{rf} \rightarrow Non\text{-}slender\ flange$		
$\lambda_w = \frac{h}{t_w} = 27.52$	Web slenderness ratio	
$\lambda_{rw} = 1.49\sqrt{\frac{E}{F_y}} = 35.884$	Limiting width to thickness parameter for a slender element subject to axial compression	(Table B4.1a, Case 5)
$\lambda_w < \lambda_{rw} \rightarrow Non\text{-}slender\ web$		
Check Effective Slenderness Ratio		
$K_y = 1$	Effective length factor for the flexural buckling about the y-axis	
$L_{by} = 0\ ft$	Distance between points which brace the member against flexural buckling about the y-axis	
$L_{cy} = K_y L_{by} = 0\ ft$	Effective length in the weak (local y) direction	(Sec. E2)
$r_y = 1.259\ in$	Radius of gyration about the local y-axis (weak)	
$K_z = 1$	Effective length factor for the flexural buckling about the z-axis	
$L_{bz} = 0\ ft$	Distance between points which brace the member against flexural buckling about the z-axis	
$L_{cz} = K_z L_{bz} = 0\ ft$	Effective length in the strong (local z) direction	(Sec. E2)
$r_z = 3.496\ in$	Radius of gyration about the local z-axis (strong)	
$E = 29000\ ksi$	Modulus of elasticity	
$F_y = 50\ ksi$	Specified minimum yield stress	
$\frac{L_{cy}}{r_y} = 0$	Effective slenderness ratio about the local y-axis (weak)	
$\frac{L_{cz}}{r_z} = 0$	Effective slenderness ratio about the local z-axis (strong)	
Check Elastic Buckling Stress		
$K_y = 1$	Effective length factor for the flexural buckling about the y-axis	
$K_z = 1$	Effective length factor for the flexural buckling about the z-axis	
$r_y = 0\ in$	Radius of gyration about the local y-axis (weak)	
$r_z = 0\ in$	Radius of gyration about the local z-axis (strong)	
$E = 29000\ ksi$	Modulus of elasticity	
$L_{by} = 0\ ft$	Distance between points which brace the member against flexural buckling about the y-axis	
$L_{cy} = K_y L_{by} = 0\ ft$	Effective length in the weak (local y) direction	(Sec. E2)
$L_{bz} = 0\ ft$	Distance between points which brace the member against flexural buckling about the z-axis	
$L_{cz} = K_z L_{bz} = 0\ ft$	Effective length in the strong (local z) direction	(Sec. E2)

$C_w = 152\ in^6$	Warping constant	
$G = 11154\ ksi$	Shear modulus of elasticity	
$J = 0.282\ in^4$	Torsional constant	
$I_y = 9.77\ in^4$	Moment of inertia about local y-axis (weak)	
$I_z = 75.3\ in^4$	Moment of inertia about local z-axis (strong)	
$F_y = 50\ ksi$	Specified minimum yield stress	
$F_e = \dfrac{\pi^2 E}{\left(\max(\frac{L_{cy}}{r_y}, \frac{L_{cz}}{r_z})\right)^2} = 2.862e+11\ ksi$	Flexural elastic buckling stress	(Eq. E3-4)
$F_n = \left(0.658^{\frac{F_y}{F_e}}\right) F_y = 50\ ksi$	Flexural nominal buckling stress	(Eq. E3-2)
Check Available Compressive Strength		
$F_{n,min} = \min(F_n, F_{n,FT}) = 50\ ksi$	Nominal stress	
$A_g = 6.16\ in^2$	Gross cross-sectional area	
$\Omega_c = 1.67$	Safety factor for compression	
$P_n = F_n A_g = 308\ k$	Nominal compressive strength - flexural buckling	(Eq. E3-1)
$P_{nc} = \dfrac{P_n}{\Omega_c} = 184.431\ k$	Available compressive strength	(Sec. E1)
$P_r(comp) = 0\ k$	Required axial compressive strength at governing location	

Flexural Analysis (Strong Axis)	37.249 k-ft	50.898 k-ft	-	-

Calculate Lateral-Torsional Buckling Factor		
$C_b = 1$	Lateral-torsional buckling modification factor	
Check Slenderness Ratio: Flexure		
$E = 29000\ ksi$	Modulus of elasticity	
$F_y = 50\ ksi$	Specified minimum yield stress	
$t_f = 0.4\ in$	Thickness of flange	
$b = \dfrac{b_f}{2} = 2.635\ in$	Half the full flange width	
$k_{des} = 0.7\ in$	Flange thickness plus fillet depth	
$h = d - 2k_{des} = 6.88\ in$	Clear depth of web inbetween flanges	
$\lambda_f = \dfrac{b}{t_f} = 6.588$	Flange slenderness ratio	
$\lambda_{rf} = 1.00\sqrt{\dfrac{E}{F_y}} = 24.083$	Limiting width to thickness parameter for a slender element	(Table B4.1b, Case 10)
$\lambda_{pf} = 0.38\sqrt{\dfrac{E}{F_y}} = 9.152$	Limiting width to thickness parameter for a compact element	(Table B4.1b, Case 10)
$\lambda_f < \lambda_{pf} \rightarrow Compact\ flange$		
$\lambda_w = \dfrac{h}{t_w} = 27.52$	Web slenderness ratio	
$\lambda_{rw} = 5.70\sqrt{\dfrac{E}{F_y}} = 137.274$	Limiting width to thickness parameter for a slender element	(Table B4.1b, Case 15)

$\lambda_{pw} = 3.76\sqrt{\frac{E}{F_y}} = 90.553$	Limiting width to thickness parameter for a compact element	(Table B4.1b, Case 15)
$\lambda_w < \lambda_{pw} \rightarrow Compact\ web$		
Check Yielding		
$F_y = 50\ ksi$	Specified minimum yield stress	
$Z_z = 20.4\ in^3$	Plastic section modulus about local z-axis (strong)	
$M_n = M_p = F_y Z_z = 85\ k\text{-}ft$	Nominal flexural strength for the plastic limit state	(Eq. F2-1)
$Limit\ state\ of\ lateral - torsional\ buckling\ does\ not\ apply$		
$Limit\ state\ of\ flange\ local\ buckling\ does\ not\ apply$		
$Limit\ state\ of\ web\ local\ buckling\ does\ not\ apply$		
Check Available Flexural Strength (Strong Axis)		
$\Omega_b = 1.67$	Safety factor for flexure	
$M_{nz} = M_n = 85\ k\text{-}ft$	Governing nominal flexural strength (strong)	
$M_{rz} = 37.249\ k\text{-}ft$	Required flexural strength about z-axis at governing location	
$M_{cz} = \frac{M_{nz}}{\Omega_b} = 50.898\ k\text{-}ft$	Available flexural strength (strong axis)	(Sec. F1)

Flexural Analysis (Weak Axis)	0 k-ft	14.197 k-ft	-	-

Check Yielding		
$F_y = 50\ ksi$	Specified minimum yield stress	
$Z_y = 5.69\ in^3$	Plastic section modulus about local y-axis (weak)	
$S_y = 3.708\ in^3$	Elastic section modulus about local y-axis (weak)	
$M_n = M_p = F_y Z_y \leq 1.6 F_y S_y = 23.708\ k\text{-}ft$	Nominal flexural strength for the plastic limit state	(Eq. F6-1)
Check Available Flexural Strength (Weak Axis)		
$\Omega_b = 1.67$	Safety factor for flexure	
$M_{ny} = M_n = 23.708\ k\text{-}ft$	Governing nominal flexural strength (weak)	
$M_{ry} = 0\ k\text{-}ft$	Required flexural strength about y-axis at governing location	
$M_{cy} = \frac{M_{ny}}{\Omega_b} = 14.197\ k\text{-}ft$	Available flexural strength (weak axis)	(Sec. F1)

Shear Analysis (Major Axis y)	12.132 k	41.4 k	**0.293**	**PASS**

$E = 29000\ ksi$	Modulus of elasticity	
$F_y = 50\ ksi$	Specified minimum yield stress	
Check Shear Strength of Web without Tension Field Action		
$d = 8.28\ in$	Full nominal depth of the section	
$t_f = 0.4\ in$	Thickness of flange	
$k_{des} = 0.7\ in$	Flange thickness plus fillet depth	
$h = d - 2k_{des} = 6.88\ in$	Clear depth of web inbetween flanges	
$t_w = 0.25\ in$	Thickness of web	

$\lambda_w = \frac{h}{t_w} = 27.52$	Web slenderness ratio	
$\leq 2.24\sqrt{\frac{E}{F_y}} = 53.946$	Shear buckling slenderness limit	(Sec. G2.1(a))
$C_{v1} = 1$	Web shear strength coefficient	(Eq. G2-2)
$A_{vy} = dt_w = 2.07\ in^2$	Effective shear area in local y-axis	
$V_{ny} = 0.6F_yA_{vy}C_{v1} = 62.1\ k$	Nominal shear strength	(Eq. G2-1)
Check Major Axis (y) Shear Unity		
$\Omega_v = 1.5$	Safety factor for shear	
$V_{cy} = \frac{V_{ny}}{\Omega_v} = 41.4\ k$	Factored strong axis shear capacity	(Sec. G1)
$V_{ry} = 12.132\ k$	Required shear strength about the y-axis at governing location (see Applied Loading - Shear + Torsion)	
$UC = \frac{V_{ry}}{V_{cy}} = 0.293$	Shear unity check in the y-axis	

Shear Analysis (Minor Axis z)	0 k	75.737 k	**0**	**PASS**

$E = 29000\ ksi$	Modulus of elasticity	
$F_y = 50\ ksi$	Specified minimum yield stress	
Check Shear Strength of Flange without Tension Field Action		
$k_v = 1.2$	Web plate shear buckling coefficient	(Sec.G6)
$b_f = 5.27\ in$	Width of flange	
$t_f = 0.4\ in$	Thickness of flange	
$b = \frac{b_f}{2} = 2.635\ in$	Half the full flange width	
$\lambda_f = \frac{b}{t_f} = 6.588$	Flange slenderness ratio	
$\leq 1.10\sqrt{\frac{k_vE}{F_y}} = 29.02$	Shear buckling slenderness limit	(Sec. G2.2(b)(i))
$C_{v2} = 1$	Web shear buckling strength coefficient	(Eq. G2-9)
$V_n = 0.6F_y2b_ft_fC_{v2} = 126.48\ k$	Nominal shear strength	(Eq. G6-1)
Check Minor Axis (z) Shear Unity		
$\Omega_v = 1.67$	Safety factor for shear	
$V_{cz} = \frac{V_{nz}}{\Omega_v} = 75.737\ k$	Factored weak axis shear capacity	(Sec. G1)
$V_{rz} = 0\ k$	Required shear strength about the z-axis at governing location (see Applied Loading - Shear + Torsion)	
$UC = \frac{V_{rz}}{V_{cz}} = 0$	Shear unity check in the z-axis	

Bending & Axial Interaction Check (UC Bending Max)	-	-	**0.732**	**PASS**

$P_r(comp) = 0\ k$	Required axial compressive strength at governing location
$P_c = 184.431\ k$	Available axial compressive strength
$M_{rz} = 37.249\ k\text{-}ft$	Required flexural strength about z-axis at governing location

$M_{cz} = 50.898\ k\text{-}ft$	Available flexural strength (strong axis)	
$M_{ry} = 0\ k\text{-}ft$	Required flexural strength about y-axis at governing location	
$M_{cy} = 14.197\ k\text{-}ft$	Available flexural strength (weak axis)	
$For\ \frac{P_r}{P_c} < 0.2, \frac{P_r}{2P_c} + \left(\frac{M_{rz}}{M_{cz}} + \frac{M_{ry}}{M_{cy}}\right) \leq 1.0 = 0.732$	Bending and Axial Interaction	(Eq. H1-1b)